

Objective:

Motivated and detail-oriented Life Science graduate with a strong background in biochemistry, molecular biology, and computer-aided drug design. Currently aspiring to pursue a PhD program in the US to advance my expertise in biomedical sciences and engage in cutting-edge research at the intersection of molecular and computational biology. Driven to strengthen both experimental and in silico skills while contributing to multidisciplinary research that promotes scientific innovation and addresses global health challenges.

Education:

2026-Present	M.Sc in Biotechnology Brac University, Dhaka, Bangladesh	Appeared
2019-2024	B.Sc in Biochemistry and Molecular Biology University of Rajshahi, Rajshahi, Bangladesh	CGPA - 3.73/4.00
2018	Higher Secondary Certificate in Science Govt. Azizul Haque College, Bogura	GPA - 4.83/5.00
2016	Secondary School Certificate in Science A.P.Bn Public School and College, Bogura	GPA - 5.00/5.00

Experience:

Research Assistant (RA) at University of Dhaka (June 2024 – Present)

Under Assistant Professor Dr. Amlan Ganguli, Department of Clinical Pharmacy and Pharmacology

Responsibilities:

- Conducting research in the field of Bioinformatics, focusing on data analysis, microarray, DEG and computational biology.
- Assisting in literature reviews, data preprocessing, statistical analysis, algorithm development.
- Collaborating on academic publications and presentations.
- Utilizing tools such as R, Python, Cytoscape, Pyrx, Excel, GEO2R, DAVID, STRING etc

Research Associate and Instructor at Panacea Research Center. (July 2024 – Present)

Responsibilities:

- Worked with a multidisciplinary team on computer-aided drug design projects.
- Coordinated with team members to ensure alignment and progress towards research goals.
- Served as an Instructor, delivering training sessions and mentoring students in bioinformatics and computer-aided drug discovery.

Research Interest

- Computational drug design
- Genomics and Transcriptomics
- Molecular modeling and simulation
- Functional Genomics and Gene Editing
- Biological Data Analytics
- Proteomics
- Database Management and Big Data Analysis
- Network Biology and Pathway Analysis
- Data Integration and Systems Biology
- Machine Learning and Artificial Intelligence

Journal Articles:

- Md Ahad Ali, **Hriddhi Sarker**, Tania Khan, Humaira Sheikh, Ahmed Saif, Farhad Bin Farid, Sadia Afrin⁶, Most. Asha Khatun, Neeraj Kumar. Multi-Omics Pan-Cancer Profiling of CDK2 and In Silico Identification of Plant-Derived Inhibitors Using Machine Learning Approaches. **RSC Advances**, 15, 36938–36968. <https://doi.org/10.1039/d5ra05535k> (Scopus indexed; Impact Factor: 4.6; CiteScore: 7.6).
- Lamia Islam, **Hriddhi Sarker**, Md Al Amin, Rahnuma Tabassum, Arpita Nandi, Tarikul Islam, Md Azad Patwary, Fairouz Ibnat, Firoz Ahmed*, Noimul Hasan Siddiquee*. Microbial biomolecule-driven identification of next-gen GSK-3 β inhibitors for brain disorders. **Computational Biology and Chemistry**, 121, 108860 (2026). <https://doi.org/10.1016/j.compbiolchem.2025.108860> (Scopus indexed; Impact Factor: 3.1; CiteScore: 4.3; **Equal contribution as first author**)

Conference / Poster Presentations:

- **Hriddhi Sarker**, Md Ahad Ali, Humaira Sheikh, Tarikul Islam, Sujoy Banik, Md. Nurul Haque Mollah
Identification of Prognostic Regulators and Novel Therapeutic Targets in Lung Cancer via Bioinformatics and Machine Learning (Poster Presentation, Presenter). 2nd International Conference on Recent Advances in Science and Technology (ICRAST 2025). DOI: <https://doi.org/10.13140/RG.2.2.19541.69605>.
- Humaira Sheikh, Md. Ahad Ali, Nur Mohammad, **Hriddhi Sarker**, Neeraj Kumar, Md. Nurul Haque Mollah
Investigating the bioactivity of natural compounds targeting hMPV infection causing viral (F) proteins: A network pharmacology approach (Poster Presentation, Co-author). International Conference on Applied Statistics and Data Science (ICASDS 2025).

Test Scores:

IELTS, overall score- 6.5 (Listening:7.5, Reading:5.5, Speaking:6.5, Writing:6.5)

Technical Skills:

- | | |
|--|--|
| • Programming Language: | R. |
| • Bioinformatics Tools: | GEO2R, DAVID, STRING, Cytoscape. |
| • Pan Cancer Analysis: | UALCAN, GEPIA2, cBIOPORTAL, TIMER2.0 |
| • Statistical and Computational Skills: | SPSS, Data Analysis, Microsoft Excel |
| • Survey Tools: | KoboToolBox, Google Forms |
| • Academic Writing Tools: | Microsoft Word, Microsoft Powerpoint, Mendeley |
| • Computer-Aided Drug Design Skills: | PyMOL, PyRx, Discovery Studio, GROMACS, Gaussian |

Soft Skills:

- | | |
|-------------------|----------------------|
| • Problem solving | • Public speaking |
| • Time management | • Critical thinking |
| • Team work | • Project management |
| • Adaptability | • Communication |

Internship:

Research Intern at **Panacea Research Center**. (July 2024 – May 2025)

Responsibilities and Achievements:

- Worked with a multidisciplinary team on computer-aided drug design projects.
- Coordinated with team members to ensure alignment and progress towards research goals.
- Maintained detailed records of experiments, protocols, and results.
- Gained proficiency in advanced bioinformatics tools and software.

Research Intern at **BioLab Bangladesh**. (June 2024 – April 2025)

Responsibilities:

- Analyzed molecular interactions and predicted binding affinities using computational tools.
- Conducted data mining and analysis to identify potential drug targets and lead compounds.
- Collaborated with chemoinformatics experts to optimize compound libraries and screening methods.
- Presented findings in virtual meetings and contributed to scientific publications and reports.

Research Intern at **CHIRAL Bangladesh**. (July 01, 2024 – September 30, 2024)

Responsibilities and Achievements:

- Gained foundational knowledge in health research methodology.
- Collected and managed research data using mobile data collection tools like Kobo Toolbox.
- Performed research data analysis with R programming, enhancing statistical and computational skills.
- Developed expertise in scientific writing and contributed to preparing a scientific paper.
- Actively participated in workshops and hands-on sessions on publishing research findings.

Research Intern at **INMAS, Rajshahi** (December 26, 2023 – January 1, 2024)

Achievements

- Gained hands-on experience in advanced diagnostic techniques such as radioimmunoassay, bone scanning, and thyroid scanning.

Extra-Curricular Activities:

- Campus Ambassador at **Plasma Cell Research Center**.
- Member at **Theatre Idea**.
- Member at **Shangshaptak Theatre**.
- Volunteer and Member of **Nobojagoron Foundation, University of Rajshahi**.
- Provided private tuition in **science subjects to high school students**

Hobbies and Interests:

- Scientific Reading and Writing.
- Team Sports.
- Programming
- Singing
- Chess

Accomplished Trainings and Workshops:

Basics of Pharmaceuticals.

Rajshahi University Biochemistry and Molecular Biology Alumni Association, Duration: 1 day, 31 May 2024

Bioinformatics for Computer-Aided Drug Design

Panacea Research Center, Duration: 1 months, (3 May 2024 – 7 June 2024)

Bone Scanning, Radioimmunoassay, Chemiluminescent Immunoassay, Thyroid Scanning, and Renogram Techniques.

Institute of Nuclear Medicine and Allied, Duration: 7 days, (26 December 2023 – 1 January 2024)

References:

Dr. Amlan Ganguly

Assistant Professor
Department of Clinical Pharmacy and Pharmacology,
Faculty of Pharmacy
University of Dhaka, Dhaka 1000, Bangladesh
Phone: +8801919466366
Email : gangulyamlan@du.ac.bd

Dr. Mohammad. Amirul Islam

Professor
Department of Biochemistry and Molecular Biology,
University of Rajshahi. Rajshahi, Bangladesh
Phone: +8801712141273
Email : maislam14@ru.ac.bd

Dr. Mohammad Shariar Shovon

Professor
Department of Biochemistry and Molecular Biology,
University of Rajshahi. Rajshahi, Bangladesh
Phone: +88017791984274
Email : shariashovon@outlook.com

Dr. Tanzima Yeasmin

Professor
Department of Biochemistry and Molecular Biology,
University of Rajshahi. Rajshahi, Bangladesh
Phone: +8801556312361
Email : yeasmin_bio@ru.ac.bd